A low-energy-dense diet adding fruit reduces weight and energy intake in women.

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This study evaluated the effect of adding fruit or oats to the diet of free-living women on energy consumption and body weight. Fruit and oat cookies had the same amount of fiber and total calories (approximately 200 kcal), but differed in energy density. We analyzed data from a clinical trial conducted in a primary care unit in Rio de Janeiro, Brazil. Forty-nine women, ages ranging from 30 to 50 years, with body mass index (BMI)>25 kg/m2, were randomly chosen to add three apples (0.63 kcal/g energy density) or three pears (0.64 kcal/g energy density) or three oat cookies (3.7 kcal/g energy density) to their usual diet for 10 weeks. Fiber composition was similar (approximately 6g). Statistical analysis of the repeated measures of dietary composition and body weight were analyzed using mixed model procedures. Results showed a significant decrease in the energy density during the follow-up (-1.23 kcal/g, p<0.04, and -1.29 kcal/g, p<0.05) for apples and pears, respectively, compared to the oat group. The energy intake also decreased significantly (-25.05 and -19.66 kcal/day) for the apple and pear group, respectively, but showed a small increase (+0.93) for the oat group. Apples and pears were also associated (p<0.001) with weight reduction (-0.93 kg for the apple and -0.84 for the pear group), whereas weight was unchanged (+0.21; p=0.35) in the oat group. Results suggest that energy densities of fruits, independent of their fiber amount can reduce energy consumption and body weight over time.